JOIV : Int. J. Inform. Visualization, 10(2) - March 2026 565-572

INTERNATIONAL JOURNAL ON INFORMATICS VISUALIZATION

journal homepage : www.joiv.org/index.php/joiv

Towards a Sustainable Future: Accurate Predictions of CO_2 Emissions Using Hybrid LSTM-GRU Models

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Abstract*—Growing carbon dioxide (CO_2) emissions are the primary driver of global climate change, creating an urgent need for robust and reliable predictive models to support effective mitigation strategies. Accurate long-term emission forecasting is essential for policymakers to design evidence-based interventions aligned with international climate commitments. This study proposes a hybrid layered deep learning framework for long-term CO_2 emission prediction up to 2030, integrating Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures to capture complex temporal dependencies and nonlinear emission patterns. Historical emissions data were obtained from the Our World in Data database and applied to the seven largest CO_2-emitting countries, representing diverse economic structures and development trajectories. Multiple hybrid configurations were evaluated, including LSTM–LSTM, GRU–GRU, LSTM–GRU, and GRU–LSTM models, and their performance was assessed using Root Mean Square Error (RMSE). The results demonstrate that hybrid architecture consistently outperforms conventional single-layer models. The GRU–LSTM model achieved the lowest RMSE for China (520.81×10^6 tons) and Japan (24.15×10^6 tons), while the GRU–GRU model performed best for India (148.78×10^6 tons). The LSTM–GRU configuration yielded superior results for Russia (73.03×10^6 tons), and the LSTM–LSTM model showed the highest accuracy for the United States (263.14×10^6 tons). These variations highlight the importance of country-specific modeling strategies when forecasting emissions. Overall, the results confirm the effectiveness of hybrid deep learning models in capturing long-term dynamics in CO_2 emissions. The proposed framework provides reliable, data-driven forecasts that support strategic climate planning and emission-reduction efforts aligned with the Paris Agreement.

Keywords—Climate change; hybrid LSTM-GRU; emission prediction; CO_2 emission; environmental sustainability.

Manuscript received 23 Nov. 2025; revised 13 Dec. 2025; accepted 16 Jan. 2025. Date of publication 31 Mar. 2026.
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I. Introduction

Carbon dioxide (CO2) emissions are a key contribution to global warming and the greenhouse effect, making global climate change one of the century's biggest issues [1]. Over the past century, human activities like the burning of fossil fuels, changes in land use, and deforestation have caused atmospheric CO2 levels to rise to previously unheard-of levels [2]. From less than 1 billion tons annually to over 35 billion tons in 2022, CO2 emissions have skyrocketed during the Industrial Revolution, with an exponential rise, particularly after 1950 [3]. Historical patterns have demonstrated the significant contribution of energy consumption, urbanization, and industrialization to the pace of emissions acceleration [4], [5], [6].

It is becoming more and more important to be able to forecast future CO2 emissions since environmental sustainability is threatened [4]. In addition to offering information about emission trends and patterns, precise forecasts are a crucial tool for policymakers in creating more potent mitigation plans [7]. Strong prediction models can promote the global shift to a sustainable low-carbon economy [8] and help guide policy actions to meet emission reduction goals, such as those outlined in the Paris Agreement [9].

The accuracy of current predictive models is still a significant obstacle, in spite of multiple attempts to forecast carbon dioxide (CO2) emissions[10]. The intricate temporal correlations in CO2 emissions data, which are impacted by variables including energy policy[11], economic trends[12], and technology advancements[13], are frequently missed by conventional statistical models and machine learning-based techniques [14]. Because of this, forecasts frequently provide a great deal of uncertainty, particularly when used at national or regional levels, where emission patterns are very changeable because of variations in economic structure, environmental regulations, and industrialization levels [15]. Additionally, the temporal behavior of CO2 emissions is often dynamic and non-linear, with interrelated elements that are challenging to predict using traditional methods[16]. Because of this complexity, more advanced methods are needed, like time series-based deep learning models, which can identify non-linear relationships in the data as well as long-term temporal patterns [17].

The main goal of this research is to create a forecast model that can accurately anticipate carbon dioxide (CO2) emissions through 2030 using a hybrid long short-term memory (LSTM) and gated recurrent unit (GRU) [13],[19]. This strategy is predicated on the hybrid model's capacity to represent non-linear complexity and long-term temporal correlations in CO_2 emission data[20], which were previously challenging to simulate using traditional methods [21]. Identifying emission trends and patterns from seven major CO2 emitters, including China, the US, and India, is another goal of this study[10]. This work offers valuable insights into the emission dynamics of these nations, which can help inform data-driven policymaking for more successful mitigation of climate change[22]. This study is anticipated to significantly contribute to global efforts to reduce greenhouse gas emissions and promote environmental sustainability by utilizing a novel combination of sophisticated predictive models and long-term prediction capabilities ten years from now (from 2022 to 2030)[23].

Using a hybrid layered LSTM-GRU model, this study presents a novel approach to predicting carbon dioxide (CO_2) emissions. The benefit of combining these two deep learning methods is that it can capture both non-linear correlations and long-term temporal patterns in emission data[24]. This method is especially intended to address the difficulties that traditional models encounter when forecasting intricate and ever-changing emission patterns [25]. This study's contribution also benefits the global community, particularly by supporting coordinated efforts to meet the 2030 Paris Agreement targets. This study supports evidence-based strategies to minimize greenhouse gas emissions and offers pertinent insights to address global climate concerns by improving environmental sustainability through sophisticated predictive models[26].

Although deep learning models like LSTM and GRU have shown promise in capturing temporal dependencies, prior studies often relied on single-architecture frameworks, which may not optimally balance short- and long-term memory retention[27]. Furthermore, recent advancements such as Transformer-based models offer strong performance in many time-series tasks but often require larger datasets and longer training times, making them less practical for limited CO_2 datasets. This study addresses the gap by proposing a hybrid LSTM-GRU model [28], which combines the long-term memory strength of LSTM with the training efficiency of GRU. This configuration enables more accurate and computationally efficient predictions for non-linear and non-stationary CO_2 emission trends.

II. Materials and Method

The steps of the study's methodology are included in this part (Figure 1).

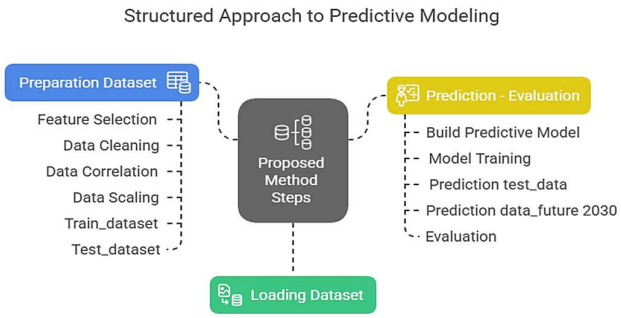


Fig. 1 Proposed method steps

A. *Theoretical Background*

This section describes the many data analysis techniques and computational methods we used in our study to forecast CO2 emissions. Both deep learning and traditional machine learning frameworks serve as the theoretical underpinnings of our approach. Because deep learning excels at processing and learning from historical datasets, it is well-suited to the complex patterns in CO2 emissions data. LSTM and GRU algorithms, which provide scalable and efficient methods for future prediction, improve deep learning.

1) Long Short-Term Memory (LSTM):

Originally known as Recurrent Neural Networks (RNN), artificial neural networks have evolved into Long Short-Term Memory (LSTM). RNN's shortcoming, however, is that it can only retain brief information from earlier data; as a result, older information will be lost (vanishing gradient) [29]. By identifying the theory that RNN uses, Hochreiter [30] and Bengio et al. [31] attempted to determine why RNN is unable to store long-term memory. As a result, RNN evolved into LSTM, which is created by adding cells to address this issue specifically, by segregating superfluous memory from unnecessary memory. Important memory will be entered into the following cell, while unnecessary memory will be eliminated [32].

The LSTM cell, Cell State (Ct), serves several vital functions, including long-term memory storage. This cell, known as the input gate (i), decides how much additional memory should be added to the cell state. This cell, known as the Forget Gate (f), determines which memory should be stored and processed to the following cell and which should be destroyed. Output Gate (o): This cell determines which memory will be hidden and which will be routed to the output. Cells Ct-1, ht-1, and Xt are the inputs carrying old information, the inputs conveying the output of the preceding cell, and the inputs carrying fresh information, respectively. While Ct is the output conveying old information to be the

input in the next cell, ht is the information generated by the cell that will be the input of the subsequent cell.

2) Gated Recurrent Unit (GRU):

The gated recurrent unit, or GRU, was developed similarly to LSTM in order to address the issue of vanishing gradients in RNN[33]. GRU differs in that it has fewer gates and a simpler cell structure [34] (Figure 2). This reduces GRU processing time by reducing the number of gates. For short-term memory, GRU has a reset gate; for long-term memory, it has an update gate

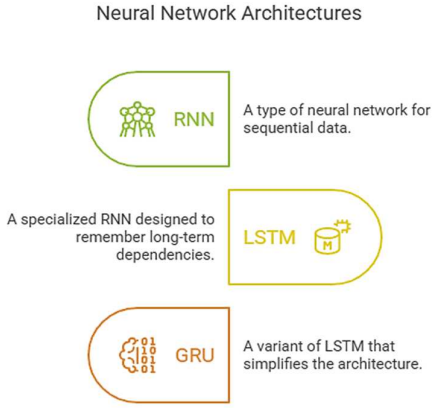


Fig. 2 RNN, LSTM, GRU architecture

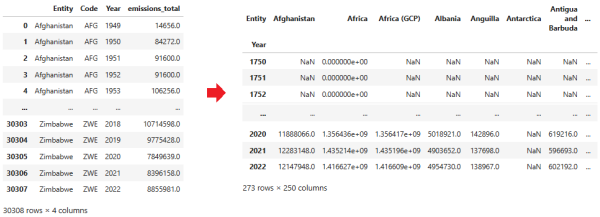
Tanh and sigmoid activation functions serve as information filters in LSTM and GRU. With a sigmoid function range of 0–1, this gate is multiplied element by element. This information is disregarded or erased if the gate value is zero; otherwise, it remains in memory. Tanh [35], which spans from 1 to +1, is a well-known non-linear activation function. The second derivative is employed to prevent information fading. Another name for the sigmoid function [36] is a non-linear activation function. The values in the sigmoid function range from 0 to 1. In essence, this function recommends which data should stay in the gate, a type of memory unit or be removed.

B. Data Source and Preprocessing

This study utilizes publicly available historical CO_2 emissions data from Our World in Data [37]**.** The dataset, which has a capacity of 850 kB (30,308 rows × 4 columns) and is in CSV format, includes 250 national names and country group names (e.g., ASEAN, EU). Key preprocessing steps included:

1) Feature Selection: Only essential columns (Year, Entity, and emissions_total) were retained for modeling.

2) Pivoting: A pivot table was applied to restructure the dataset, setting Year as the index and each selected country as a column for multivariate alignment (Figure 3).



	Entity	Code	Year	emissions_total
0	Afghanistan	AFG	1949	14656.0
1	Afghanistan	AFG	1950	84272.0
2	Afghanistan	AFG	1951	91600.0
3	Afghanistan	AFG	1952	91600.0
4	Afghanistan	AFG	1953	106256.0
...	...	...	...	...
30303	Zimbabwe	ZWE	2018	10714598.0
30304	Zimbabwe	ZWE	2019	9775428.0
30305	Zimbabwe	ZWE	2020	7849639.0
30306	Zimbabwe	ZWE	2021	8396158.0
30307	Zimbabwe	ZWE	2022	8855981.0

30308 rows × 4 columns

Entity	Afghanistan	Africa	Africa (GCP)	Albania	Anguilla	Antarctica	Antigua and Barbuda	...
Year								
1750	NaN	0.000000e+00	NaN	NaN	NaN	NaN	NaN	...
1751	NaN	0.000000e+00	NaN	NaN	NaN	NaN	NaN	...
1752	NaN	0.000000e+00	NaN	NaN	NaN	NaN	NaN	...
...	...	...	...	...	...	...	...	...
2020	11888066.0	1.356436e+09	1.356417e+09	5018921.0	142896.0	NaN	619216.0	...
2021	12283148.0	1.435214e+09	1.435196e+09	4903652.0	137698.0	NaN	596693.0	...
2022	12147948.0	1.416627e+09	1.416609e+09	4954730.0	138967.0	NaN	602192.0	...

273 rows × 250 columns

Fig. 3 Pivoting table process

3) Filtering: Filtering is done for data from countries that generated more than 1.0e9 tons of CO2 emissions in 2022, specifically China, India, Japan, Russia, and the United States, because this study will examine the impact of the nations that contribute the biggest emissions (Figures 4a and 4b).

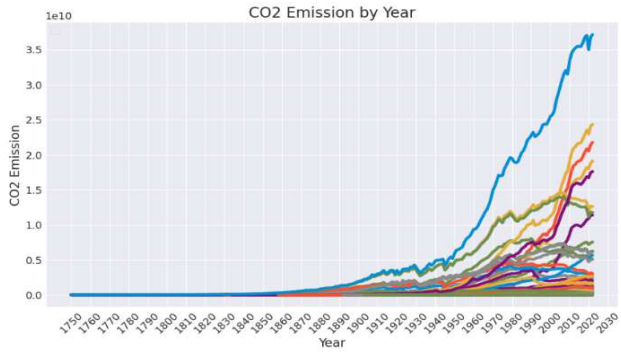


Fig. 4.a CO2 emissions from all countries in the world.

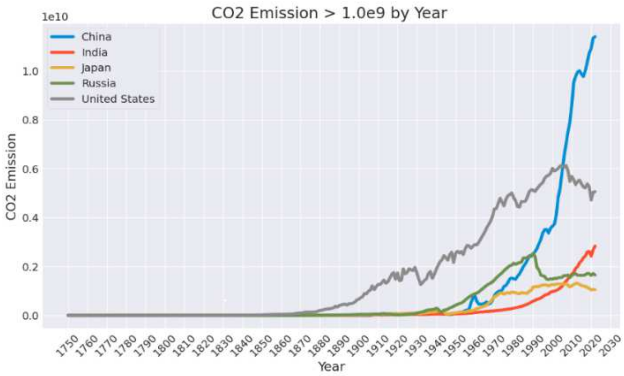


Fig. 4.b CO2 emissions from 5 countries with emissions of more than 1.0e9 tons

4) Missing Values: Emission data missing for earlier years were replaced with zeros using the .fillna(0) function in the pandas library. The missing emission values for earlier years were replaced with zero (.fillna(0) in Pandas). This decision was based on exploratory analysis indicating that interpolation produced artificially inflated emission trends in countries with sporadically available historical data. Since most missing values occurred in early years when industrial emissions were plausibly minimal or unrecorded, zero imputation was deemed a conservative yet contextually reasonable choice. Nevertheless, future work could explore spline or polynomial interpolation techniques to enhance temporal continuity without distorting trend patterns.

5) Correlation Analysis: The .corr() function was employed to analyze inter-country emission trends, followed by heatmap visualization. The.corr() method is a component of pandas, just like.fillna(). A heatmap of the correlation results for the data used in this investigation is shown in Figure 5. China and India have the strongest relationship, with a positive correlation coefficient of 0.04, suggesting that both variables are increasing. China and Russia have the poorest relationship, with a negative score of 0.0032, indicating that emissions in China have increased, whereas in Russia they have decreased.

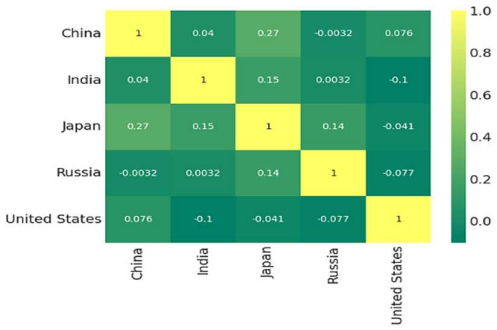


Fig. 5 Heatmap matrix

6) Normalization: Comparisons are challenging due to the data's extremely wide range of values. Standardizing data scaling to values that are easier to compare is required to address this issue. The StandardScaler() method, which is part of the Python sklearn modules, is used to accomplish this scaling. Data separation is the next phase. Figure 4 shows that, after the pivot table is completed, there are 273 rows of CO2 emission data for each country. Five rows of the dataset's 273 rows are utilized for testing, whereas 168 rows are used for training (Figure 6).

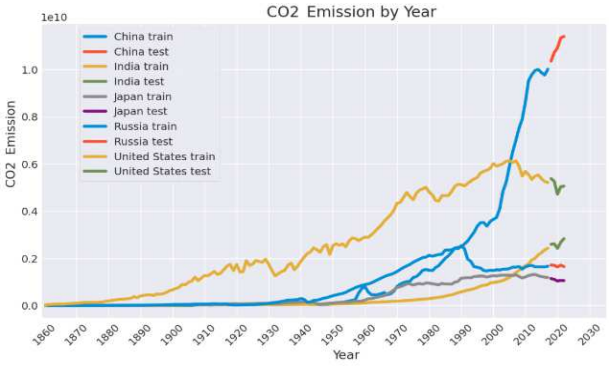


Fig. 6 Graph of splitting results for each country

C. Model Design and Architecture

To effectively model sequential and nonlinear trends in CO_2 emissions, we constructed a deep learning architecture that combines LSTM and GRU architectures. Four model variants were developed: LSTM–LSTM (LL), GRU–GRU (GG), LSTM–GRU (LG), GRU–LSTM (GL).
Each model follows a common architecture layout:

1) Input Layer: Accepts time-series sequences with a shape of (batch_size=50, time_steps=3, features=1).

2) Two Recurrent Layers: Depending on the model configuration, two sequential LSTM or GRU layers with 128 hidden units each are stacked.

3) Fully Connected Layers: A dense layer with 25 neurons applies a non-linear transformation, followed by a final dense layer with 1 neuron to predict the next CO_2 emission value.

The architecture was implemented in TensorFlow using the Keras API for model prototyping, visualization, and training.

D. Training and Evaluation

The training data must be transformed (reshaped) into a 3-dimensional array because it is now in the form of a 2-dimensional array. A batch size of 128, a timestep comprising three emission data points for each country in sequential order, and a feature with a value of 1 are also required, as the input for prediction consists solely of data with the attribute "emission_total." LSTM-LSTM (LL), GRU-GRU (GG), GRU-LSTM (GL), and LSTM-GRU (LG) are the orders of the four hidden layers in the model architecture. The first two layers use layer variations (LSTM and GRU). For the first two hidden layers, each hidden layer employs a variable of 128 neurons. The final layer consists of two dense layers, each containing one neuron and twenty-five neurons. Only one summary model, LSTM-GRU (LG), is shown to conserve space (Figure 7).

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 32, 128)	66,560
gru (GRU)	(None, 128)	99,072
dense (Dense)	(None, 25)	3,225
Dense_1 (Dense)	(None, 1)	26

Total params: 168,883 (659.70 KB)
Trainable params: 168,883 (659.70 KB)
Non-trainable params: 0 (0.00 B)

Fig. 7 Summary for the LG model

The first layer (LSTM or GRU, depending on the architecture) will use 128 neurons to train the train_dataset, as shown in Figure 8. Using 128 neurons, the first layer's output will be sent to the second layer (LSTM or GRU). The third layer, which is dense and has 25 neurons, will receive the output from the second layer, and the last layer is dense and has one neuron.

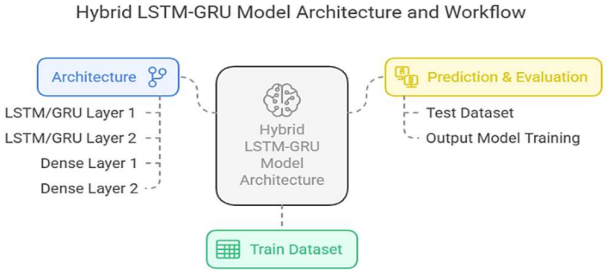


Fig. 8 Hybrid LSTM-GRU prediction model architecture

Learning_rate=0.005, epoch=500, and loss evaluation for the training process utilizing MSE (Mean Squared Error) and MAE (Mean absolute error) metrics are used for model compilation using the RMSprop optimizer. When evaluating test data, RMSE (Root Mean Squared Error) and R-squared (Coefficient of Determination) are used in addition to MSE and MAE measurements. The average difference between predicted and actual values is measured by MAE. The average squared difference between predicted and actual values is measured by MSE. The root value of MSE is called RMSE. If the metric's error rate is lower, the prediction results will be more accurate. Predictions pertaining to time series are frequently assessed using these metrics [37]. The goodness-of-fit of the model to the data is quantified by the R-squared statistic. The four measurements can be expressed as mathematical equations.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|Y_i - \hat{y}|^2 \quad (1)$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{y})^2 \quad (2)$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{y})^2} \quad (3)$$

$$R^2 = 1 - \frac{\sum(Y_i - \hat{y})^2}{\sum(Y_i - \bar{y})^2} \quad (4)$$

where,

$\hat{y} = predicted\ value\ of\ y$
$\bar{y} = mean\ value\ of\ y$

The model trained on this data can be used to predict a range of data types. MSE and MAE are used to evaluate training results. Test_dataset is used for model testing, prediction generation, and performance evaluation using RMSE, MSE, R, and MAE. With unknown input data, this training model can also be used to make future predictions. Although the available dataset extends only to 2022, projections will be made through 2030, as in this study. As is well known, GRU and LSTM models can only forecast one data point given the previous dataset. Since the input data is unavailable, it is not possible to predict the second dataset after the last dataset. The initial prediction results will be used as input to a new dataset, enabling predictions to be produced up to a few steps beyond the final dataset. In order to generate a second prediction, a new dataset input (original dataset + first prediction results) can be used. The third prediction then uses the second data prediction as input to a new dataset. Until the projection approaches 2030, this pattern is repeated (Figure 9).

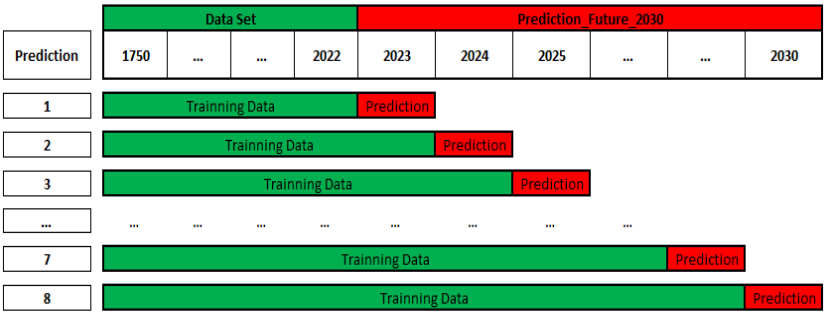
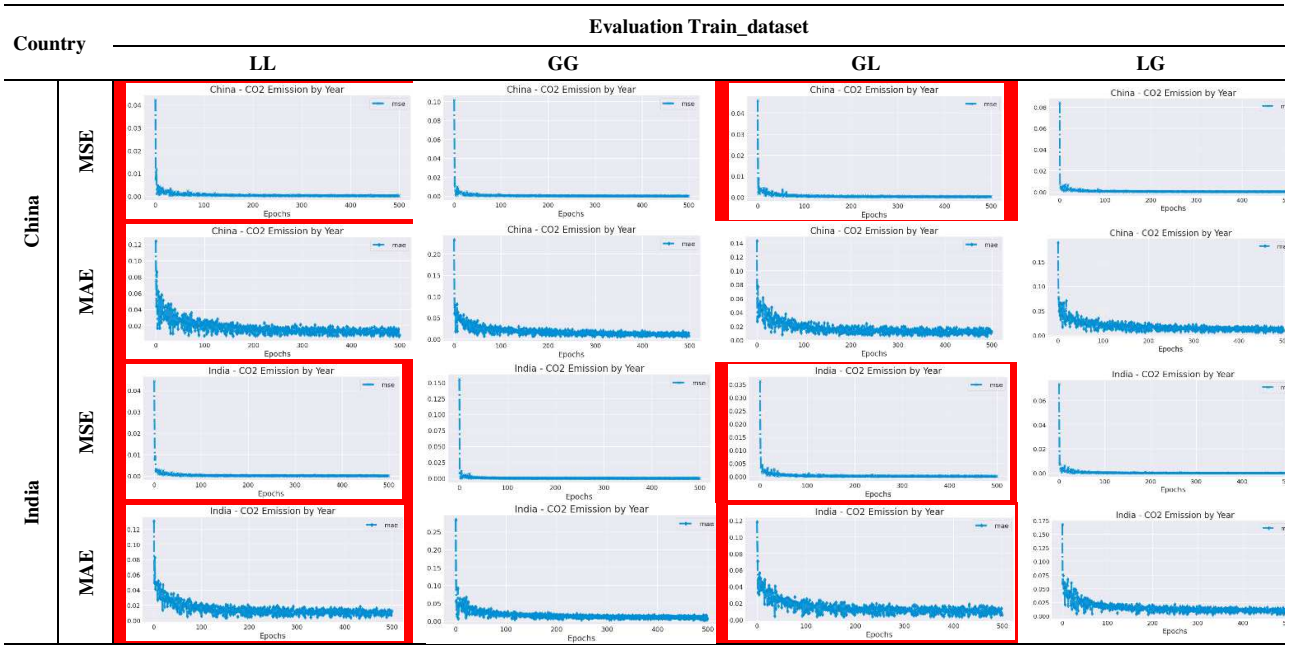


Fig. 9 Use of input data for long-term predictions up to 2030

III. Results and Discussion

MSE and MAE are used to measure each epoch of the 500-epoch training process using the train_dataset for each country, and the results are shown graphically (Figure 10). The LSTM-LSTM (LL) model typically yields lower loss estimates for the US, Russia, and Japan than other models. Nonetheless, the GRU-LSTM (GL) and LL models yield nearly identical loss estimates for China and India.



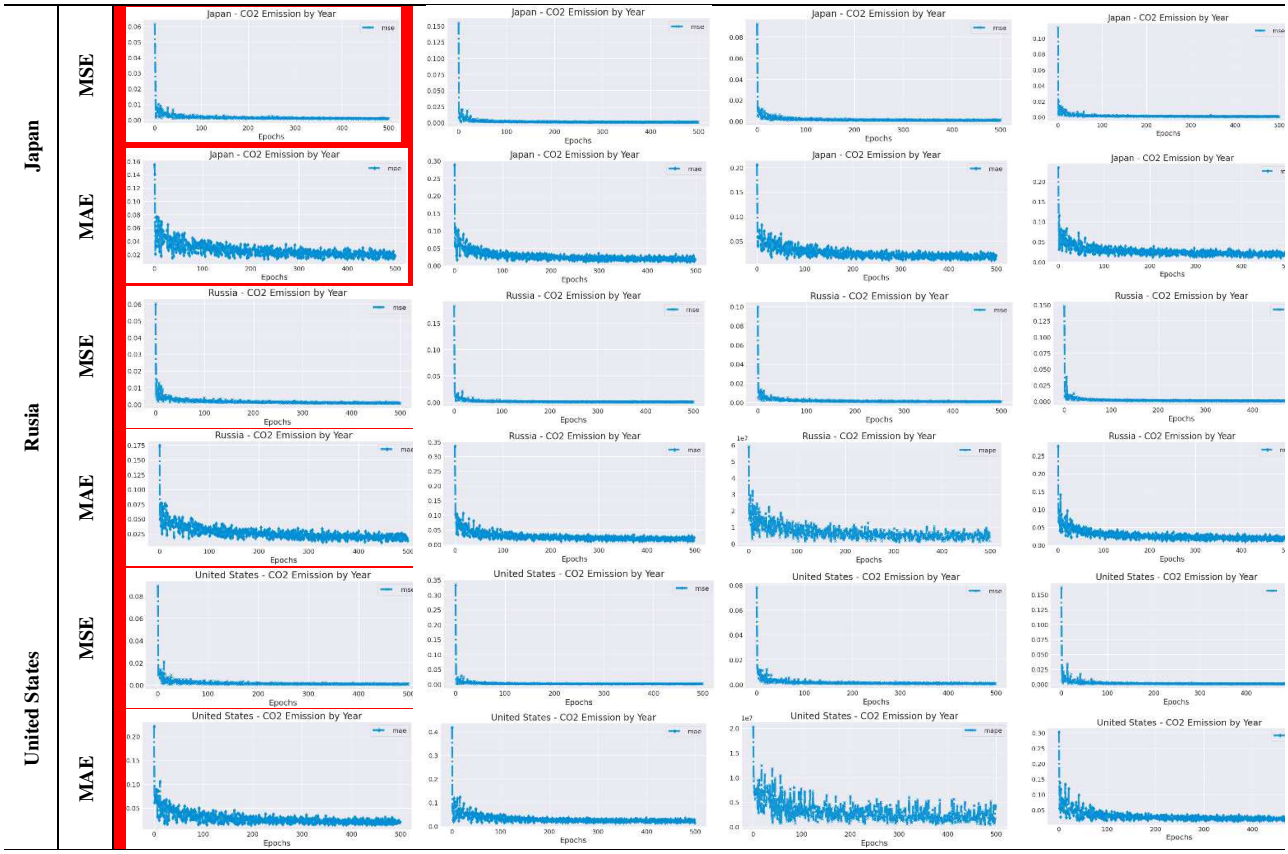


Fig. 10 Results of MAE-MSE loss measurements using train_dataset

The test_dataset is used to evaluate each model produced during training. One table (Figure 11) summarizes the loss computations for MAE, MSE, and RMSE. For China, India, Japan, Russia, and the US, the models with the lowest loss values are GL, GG, GL, LG, and LL, respectively. Japan had the highest match value with the real data (positive R value) among the five countries.

Country	Model	MAE (e6)	MSE (e15)	RMSE (e6)	R
China	LL	546,65	366,50	605,40	-1.42
	GG	1.055,09	1.180,66	1.086,58	-6.79
	GL	455,25	271,25	520,81	-0.79
	LG	1.016,47	1.103,27	1.050,37	-6.28
India	LL	115,90	26,32	162,25	-0.52
	GG	96,96	22,13	148,78	-0.27
	GL	145,74	24,07	155,13	-0.39
	LG	123,05	28,27	168,13	-0.63
Japan	LL	21,29	0,61	24,65	0.58
	GG	71,26	5,71	75,60	-3.07
	GL	17,86	0,58	24,15	0.57
	LG	29,11	1,09	33,09	0.22
Rusia	LL	80,78	11,27	106,16	-8.97
	GG	68,91	6,04	77,75	-4.35
	GL	180,27	38,02	194,99	-32.63
	LG	59,87	5,33	73,03	-3.72
United States	LL	157,14	69,24	263,14	-0.34
	GG	213,18	112,92	336,04	-1.19
	GL	182,73	77,60	278,57	-0.51
	LG	192,26	95,18	308,52	-0.85

Fig. 11 Results of MAE, MSE, RMSE, R loss measurements using the test_dataset

Using four models (LL, GG, GL, and LG), the outcomes of long-term projections for 2022 to 2030, without data, are presented as graphs for each nation (Figure 12). The model with the lowest loss, previously obtained on the test_dataset, is used to select the best-performing graph. China and Japan have the best GL model for estimating emissions, followed by India (GG), Russia (LG), and the United States (LL). The absence of actual data for the 2022–2030 period precludes calculating losses for this long-term forecast. According to the best model for each nation's projections through 2030, China, India, and Japan do not exhibit rising emissions (the slope remains flat). In contrast, the US and Russia have experienced substantial growth (the slope is increasing).

The variation in model performance across countries likely stems from differences in the characteristics of emission trends. For instance, the GRU-LSTM model performed best in China and Japan, possibly because these countries exhibit mixed short-term fluctuations and long-term structural changes, which are well suited to the GRU's efficiency and the LSTM's long-memory modeling. In contrast, Russia's emission data showed a smoother long-term trend, which favored the LSTM-GRU sequence. India's performance with GRU-GRU may indicate that its emission series are governed by shorter-term shocks, such as policy shifts or energy crises, where GRU's simplicity offers an advantage. This heterogeneous pattern underscores the importance of customizing deep learning architectures based on country-specific emission dynamics.

Despite the promising results, this study has several limitations. First, the model relies solely on historical total CO_2 emissions without incorporating sectoral or economic activity variables, which may limit its explanatory power. Second, imputing missing data with zeros, while practical,

could introduce biases, particularly in countries with sparse records. Third, the hybrid model does not currently leverage attention mechanisms, which may capture complex dependencies more effectively. Additionally, the forecasting approach assumes that emission drivers remain relatively stable, which may not hold in the face of sudden geopolitical or technological shifts. Future work should address these limitations by integrating multi-source datasets and advanced temporal modeling techniques such as Transformers or attention-based LSTM variants.

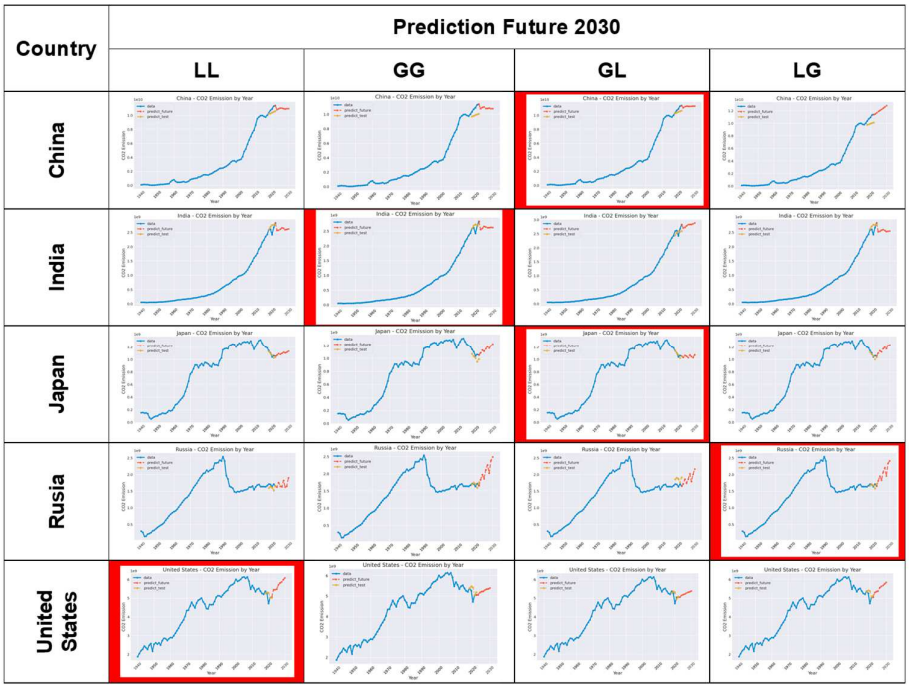


Fig. 12 Prediction result graph until 2030

IV. Conclusion

This paper proposes two methods for achieving high-accuracy CO2 emission projections for the top five CO2-emitting nations. The first method employs hybrid LSTM-GRU architecture to select the best-performing model from a two-layer architecture. Meanwhile, the second method extends predictions beyond the available dataset (which ends in 2022) by iteratively using forecasted results as new input data, allowing long-term projections until 2030. The study finds that different nations benefit from distinct model architectures: China and Japan favor GRU-LSTM, India uses GRU-GRU, Russia adopts LSTM-GRU, and the US employs LSTM-LSTM to achieve the lowest prediction loss.

The long-term projections suggest that China, India, and Japan will maintain stable emissions trends until 2030, while the US and Russia are expected to experience significant increases. By focusing on these major contributors to global CO2 emissions, the study provides valuable insights that enable more data-driven mitigation strategies. Moreover, the application of deep learning models underscores their potential in shaping effective environmental policies aligned with global sustainability goals, such as the Paris Agreement. The development of this hybrid predictive framework also paves the way for future research in complex environmental forecasting, offering a robust and scalable method for addressing pressing climate challenges.

This study opens several pathways for advancing CO2 emission prediction and climate change mitigation. Future research could enhance the hybrid LSTM-GRU model by integrating advanced deep learning techniques such as Transformers or Attention Mechanisms to improve accuracy and capture complex emission patterns. Expanding the dataset with sector-specific emissions, socioeconomic factors, and policy changes would increase its applicability. Additionally, shifting the focus to regional or local analyses could provide more precise mitigation strategies. Integrating this CO2 emission forecast model with global climate models would help assess the impact of emission variations on temperature rise, glacier melting, and extreme weather events. Developing interactive tools based on this model could assist policymakers in simulating mitigation scenarios and evaluating real-time policy implications. Furthermore, extending the model to assess the effectiveness of sustainability policies, such as renewable energy transitions and low-carbon technologies, would enhance its real-world relevance. By leveraging emerging technologies and richer datasets, future research can refine CO2 emission predictions and strengthen global efforts to combat climate change.

Acknowledgment

This research was conducted without any external funding or financial support from third-party organizations. All resources, including data collection, model development, and analysis, were carried out independently by the research team. The absence of external funding ensures the study's impartiality and enables an objective examination of CO_2 emission projections. Despite the lack of financial sponsorship, the research remains committed to contributing valuable insights to the global effort to mitigate climate

change through the application of advanced deep learning techniques.

REFERENCES

[1] IPCC, "Climate Change 2021: The Physical Science Basis," Intergovernmental Panel on Climate Change, Geneva, Switzerland, 2021. [Online]. Available: https://www.ipcc.ch/report/ar6/wg1/.

[2] H. Ritchie, P. Rosado, and M. Roser, "CO_2 and Greenhouse Gas Emissions," *Our World in Data*, 2023. [Online]. Available: https://ourworldindata.org/co2-and-greenhouse-gas-emissions.

[3] P. Friedlingstein et al., "Global Carbon Budget 2021," *Earth Syst. Sci. Data*, 2021. [Online]. Available: https://www.earth-system-science-data.net/about/news_and_press/2021-11-04_global-carbon-budget-2021.html.

[4] R. Qiao et al., "Industrialization, urbanization, and innovation: Nonlinear drivers of carbon emissions in Chinese cities," *Appl. Energy*, vol. 358, p. 122598, Mar. 2024, doi:10.1016/j.apenergy.2023.122598.

[5] J. C. Onwe, E. Ullah, M. A. Ansari, M. Sahoo, and K. S. Dhayal, "Industrialization meets sustainability: Analysing the role of technological innovations, energy efficiency and urbanisation for major industrialized economies," *J. Environ. Manag.*, vol. 372, p. 123297, Dec. 2024, doi: 10.1016/j.jenvman.2024.123297.

[6] Sumaira and H. M. A. Siddique, "Industrialization, energy consumption, and environmental pollution: evidence from South Asia," *Environ. Sci. Pollut. Res.*, vol. 30, no. 2, pp. 4094–4102, Aug. 2022, doi: 10.1007/s11356-022-22317-0.

[7] C. Le Quéré et al., "Global Carbon Budget 2020," Global Carbon Project, 2020. [Online]. Available: https://www.globalcarbonproject.org/global/images/carbonbudget/Infographic_Emissions2020.pdf.

[8] J. Luo, W. Zhuo, S. Liu, and B. Xu, "The Optimization of Carbon Emission Prediction in Low Carbon Energy Economy Under Big Data," *IEEE Access*, vol. 12, pp. 14690–14702, 2024, doi:10.1109/access.2024.3351468.

[9] A. M. Begum and M. A. Mobin, "A machine learning approach to carbon emissions prediction of the top eleven emitters by 2030 and their prospects for meeting Paris agreement targets," *Sci. Rep.*, vol. 15, no. 1, Jun. 2025, doi: 10.1038/s41598-025-04236-5.

[10] J. Hua et al., "Artificial intelligence for calculating and predicting building carbon emissions: a review," *Environ. Chem. Lett.*, vol. 23, no. 3, pp. 783–816, Mar. 2025, doi: 10.1007/s10311-024-01799-z.

[11] T. Grodzicki and M. Jankiewicz, "The impact of renewable energy and urbanization on CO2 emissions in Europe – Spatio-temporal approach," *Environ. Dev.*, vol. 44, p. 100755, Dec. 2022, doi:10.1016/j.envdev.2022.100755.

[12] P. Mitić, A. Fedajev, M. Radulescu, and A. Rehman, "The relationship between CO2 emissions, economic growth, available energy, and employment in SEE countries," *Environ. Sci. Pollut. Res.*, vol. 30, no. 6, pp. 16140–16155, Sep. 2022, doi: 10.1007/s11356-022-23356-3.

[13] H. Chen, J. Yi, A. Chen, D. Peng, and J. Yang, "Green technology innovation and CO2 emission in China: Evidence from a spatial-temporal analysis and a nonlinear spatial durbin model," *Energy Policy*, vol. 172, p. 113338, Jan. 2023, doi:10.1016/j.enpol.2022.113338.

[14] D. Huo et al., "Near-real-time daily estimates of fossil fuel CO2 emissions from major high-emission cities in China," *Sci. Data*, vol. 9, no. 1, Nov. 2022, doi: 10.1038/s41597-022-01796-3.

[15] M. Mondal, "Staving off Climate Change for Assuring Sustainable Development: India's Domestic Policies Amidst International Climate Dialogues," in *Climate Change and Regional Socio-Economic Systems in the Global South*, M. Mishra, A. J. de Lucena, and B. Maharaj, Eds., Singapore: Springer, 2024, doi: 10.1007/978-981-97-3870-0_15.

[16] N. Ahmed, G. Xinagyu, M. Alnafissa, A. Ali, and H. Ullah, "Linear and non-linear impact of key agricultural components on greenhouse gas emissions," *Sci. Rep.*, vol. 15, no. 1, Feb. 2025, doi:10.1038/s41598-025-88159-1.

[17] R. Jaiswal, G. K. Jha, R. R. Kumar, and K. Choudhary, "Deep long short-term memory based model for agricultural price forecasting," *Neural Comput. Appl.*, vol. 34, no. 6, pp. 4661–4676, Nov. 2021, doi: 10.1007/s00521-021-06621-3.

[18] N. Aini, W. Hastomo, and R. Y. Go, "Prediction of anthropogenic greenhouse gas emissions via manure management in Indonesia and alternative policies for Indonesian livestock development," J. Renew. Energy Environ., vol. 10, no. 3, pp. 99–106, 2023.

[19] W. Hastomo, N. Aini, A. S. B. Karno, and L. M. R. Rere, "Metode Pembelajaran Mesin untuk Memprediksi Emisi Manure Management," *J. Nas. Tek. Elektro Teknol. Inf.*, vol. 11, no. 2, pp. 131–139, May 2022, doi: 10.22146/jnteti.v11i2.2586.

[20] B. Wu, H. Zeng, Z. Wang, and L. Wang, "Interpretable short-term carbon dioxide emissions forecasting based on flexible two-stage decomposition and temporal fusion transformers," *Appl. Soft Comput.*, vol. 159, p. 111639, Jul. 2024, doi: 10.1016/j.asoc.2024.111639.

[21] L. Wartschinski, Y. Noller, T. Vogel, T. Kehrer, and L. Grunske, "VUDENC: Vulnerability Detection with Deep Learning on a Natural Codebase for Python," Inf. Softw. Technol., vol. 144, p. 106809, Apr. 2022, doi: 10.1016/j.infsof.2021.106809.

[22] C. Yang, S. Bu, Y. Fan, W. X. Wan, R. Wang, and A. Foley, "Data-driven prediction and evaluation on future impact of energy transition policies in smart regions," *Appl. Energy*, vol. 332, p. 120523, Feb. 2023, doi: 10.1016/j.apenergy.2022.120523.

[23] G. M. I. Alam et al., "Deep learning model based prediction of vehicle CO2 emissions with eXplainable AI integration for sustainable environment," *Sci. Rep.*, vol. 15, no. 1, Jan. 2025, doi:10.1038/s41598-025-87233-y.

[24] D. Sharma, S. Thapar, A. Masood, and K. Sachdeva, "Fourier-Enhanced Deep Learning and Machine Learning Models for Predicting Multi-Scale PM2.5 Dynamics in Megacities: A Case Study of Delhi," *Earth Syst. Environ.*, Sep. 2025, doi: 10.1007/s41748-025-00778-w.

[25] Y. Xiao et al., "A Survey on Non-Autoregressive Generation for Neural Machine Translation and Beyond," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 11407–11427, Oct. 2023, doi:10.1109/tpami.2023.3277122.

[26] S. Toumi, A. Aljadani, H. Toumi, B. Ammouri, and M. Dhiabi, "AI for climate change: unveiling pathways to sustainable development through greenhouse gas emission predictions," *Eurasian Econ. Rev.*, vol. 15, no. 4, pp. 923–963, Jan. 2025, doi: 10.1007/s40822-024-00295-7.

[27] D. Shen, X. Tao, V. Koncar, and J. Wang, "A hybrid deep learning framework combined with intelligent garment system for real-time fatigue monitoring in cycling," *AATCC J. Res.*, vol. 12, no. 4, Jul. 2025, doi: 10.1177/24723444251353784.

[28] Y. Sujatna et al., "Stacked LSTM-GRU Long-Term Forecasting Model for Indonesian Islamic Banks," *Knowl. Eng. Data Sci.*, vol. 6, no. 2, p. 215, Nov. 2023, doi: 10.17977/um018v6i22023p215-250.

[29] B. Yang, F. Fan, R. Ni, J. Li, L. Kiong, and X. Liu, "Continual learning-based trajectory prediction with memory augmented networks," *Knowl.-Based Syst.*, vol. 258, p. 110022, Dec. 2022, doi:10.1016/j.knosys.2022.110022.

[30] R. Dornberger, "New Trends in Business Information Systems and Technology," *Springer*, vol. 294, no. 1, pp. 50–312, 2019.

[31] C. Li, F. Yang, and J. Yang, "The Role of Long-Term Dependency in Synthetic Speech Detection," *IEEE Signal Process. Lett.*, vol. 29, pp. 1142–1146, 2022, doi: 10.1109/lsp.2022.3169954.

[32] J. Oruh, S. Viriri, and A. Adegun, "Long Short-Term Memory Recurrent Neural Network for Automatic Speech Recognition," *IEEE Access*, vol. 10, pp. 30069–30079, 2022, doi:10.1109/access.2022.3159339.

[33] Y. Wu, F. Hu, C. Yue, and S. Sun, "Residual-time gated recurrent unit," *Neurocomputing*, vol. 624, p. 129396, Apr. 2025, doi:10.1016/j.neucom.2025.129396.

[34] T. Das and S. Guchhait, "A hybrid GRU and LSTM-based deep learning approach for multiclass structural damage identification using dynamic acceleration data," *Eng. Fail. Anal.*, vol. 170, p. 109259, Mar. 2025, doi: 10.1016/j.engfailanal.2024.109259.

[35] H. U. Rehman, A. U. Awan, E. M. Tag-ElDin, S. E. Alhazmi, M. F. Yassen, and R. Haider, "Extended hyperbolic function method for the (2+1)-dimensional nonlinear soliton equation," *Results Phys.*, vol. 40, p. 105802, Sep. 2022, doi: 10.1016/j.rinp.2022.105802.

[36] L. Wang, W. Ye, Y. Zhu, F. Yang, and Y. Zhou, "Optimal parameters selection of back propagation algorithm in the feedforward neural network," *Eng. Anal. Bound. Elem.*, vol. 151, pp. 575–596, Jun. 2023, doi: 10.1016/j.enganabound.2023.03.033.

[37] H. Hewamalage, K. Ackermann, and C. Bergmeir, "Forecast evaluation for data scientists: common pitfalls and best practices," *Data Min. Knowl. Discov.*, vol. 37, no. 2, pp. 788–832, Dec. 2022, doi: 10.1007/s10618-022-00894-5.